LESSON 7.8

Ordered Pairs in All Four Quadrants

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can plot ordered pairs in all four quadrants of the coordinate plane.

WHY THIS MATTERS Captain Vega realized her treasure map only covered one corner of the island! She needs to expand the map in all directions — north, south, east, and west of the base camp at the origin (0, 0). Now clues can have negative coordinates: (-4, 3) means 4 units west and 3 units north. A clue at (2, -5) means 2 units east and 5 units south. Which quadrant is each clue in?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Quadrant *Spanish: Cuadrante*	One of the four parts of a coordinate grid, numbered I to IV.	I (+, +) top-right, II (-, +) top-left, III (-, -) bottom-left, IV (+, -) bottom-right
Negative coordinate *Spanish: Coordenada negativa*	A coordinate less than zero. It means left or below the center point.	(-3, 2) means 3 left and 2 up from the origin
Reflection *Spanish: Reflexión*	A flipped, mirror image across a line.	(3, 2) reflected over the y-axis is (-3, 2) — same distance, opposite side
Axis *Spanish: Eje*	One of the two number lines that frame a graph: the x-axis runs across, the y-axis runs up and down.	The x-axis goes left-right; the y-axis goes up-down; they cross at (0, 0)
Integer *Spanish: Número entero*	Whole numbers and their opposites, like -2, -1, 0, 1, 2.	..., -3, -2, -1, 0, 1, 2, 3, ...

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

I will plot (-3, -4). I start at the origin (0, 0).

1. The x is -3, which is negative, so I move 3 units to the LEFT.
2. The y is -4, which is negative, so I move 4 units DOWN. I place my dot.
3. Both coordinates are negative, so the point is in Quadrant III.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now let's plot (-5, 2). The x is -5, so which way do we move? Left 5 units.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- In Quadrant _____, the x-coordinate is _____ and the y-coordinate is _____. I know (-3, -4) is in Quadrant _____ because both coordinates are _____.

Word bank Quadrant | Negative coordinate | Reflection | Axis | Integer | quadrant | positive | negative | x-coordinate | y-coordinate | left | right

WATCH OUT!

A common mistake

A common mistake in Ordered Pairs in All Four Quadrants is matching each sign to the wrong axis

LESSON 7.8

Ordered Pairs in All Four Quadrants

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can plot ordered pairs in all four quadrants of the coordinate plane.

WHY THIS MATTERS Captain Vega realized her treasure map only covered one corner of the island! She needs to expand the map in all directions — north, south, east, and west of the base camp at the origin (0, 0). Now clues can have negative coordinates: (-4, 3) means 4 units west and 3 units north. A clue at (2, -5) means 2 units east and 5 units south. Which quadrant is each clue in?

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

I will plot (-3, -4). I start at the origin (0, 0).

1. The x is -3, which is negative, so I move 3 units to the LEFT.
2. The y is -4, which is negative, so I move 4 units DOWN. I place my dot.
3. Both coordinates are negative, so the point is in Quadrant III.

PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each ordered pair to its quadrant.

_____ (3, 4)	(A) Quadrant I
_____ (-2, 5)	(B) Quadrant II
_____ (-6, -1)	(C) Quadrant III
_____ (1, -3)	(D) Quadrant IV
_____ (-4, 2)	(E) Quadrant II
_____ (5, -6)	(F) Quadrant IV

STEPS TO TRY

1. Only the SIGNS matter: (+, +), (-, +), (-, -), and (+, -) each belong to one quadrant.
2. Use the sign map: I = (+, +), II = (-, +), III = (-, -), IV = (+, -). Read each pair's signs and look them up.
3. Start with (3, 4): both positive, so Quadrant I. Now read the signs of (-2, 5) and use the map.

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

In which quadrant is the point (-4, 5)?

(A) Quadrant I

(B) Quadrant III

(C) Quadrant II

(D) Quadrant IV

STEPS TO TRY

1. Look at the signs: x = -4 is negative and y = 5 is positive.
2. Negative x means move LEFT; positive y means move UP. Ask which corner of the plane is left-and-up.
3. From the origin, slide left 4 and climb up 5. Quadrants count I, II, III, IV counterclockwise from top-right.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A point has a positive x-coordinate and a negative y-coordinate. Which quadrant is it in?

(A) Quadrant I

(B) Quadrant II

(C) Quadrant IV

(D) Quadrant III

STEPS TO TRY

1. Positive x pushes you RIGHT; negative y drops you DOWN.
2. Find the corner that is right of the y-axis and below the x-axis. Quadrants I–IV go counterclockwise from top-right.
3. Try a sample point like (3, -2): go right 3, then down 2. See which corner of the grid that lands in.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Which point is located on the x-axis?

(A) (-4, -4)

(B) (0, 4)

(C) (4, 0)

(D) (4, 4)

STEPS TO TRY

1. A point ON the x-axis never moves up or down — its height is zero.
2. Check each pair's y-coordinate (the second number). On the x-axis, the y must be 0.
3. Read the second number in each pair. A point sits on the x-axis only when that number is 0.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How do the signs of the x- and y-coordinates tell you which quadrant a point is in?

SENTENCE STARTER In Quadrant ___, the x-coordinate is ___ and the y-coordinate is ___. I know (-3, -4) is in Quadrant ___ because both coordinates are ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help